

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 01

1) Hello World Program

Description: Write a simple Java program that prints "Hello, World!" to the console.

Hint: Use the `System.out.println()` method to print the text.

2) Addition of Two Numbers

Description: Write a Java program to add two numbers entered by the user and display the result.

Hint:

- Use the `Scanner` class to take input from the user.
- Sum the two numbers using the `+` operator and display the result using `System.out.println()`.

3) Find Maximum of Three Numbers

Description: Write a Java program to find the maximum of three numbers entered by the user.

Hint:

- Use if-else statements or the ternary operator `?:` to compare the numbers.
- Consider using nested `if` statements for clarity.

4) Check Even or Odd Number

Description: Write a Java program that checks if a number entered by the user is even or odd.

Hint:

- Use the modulus operator `%` to determine if the number is even (`number % 2 == 0`) or odd (`number % 2 != 0`).
- Display the result using `System.out.println()`.

5) Factorial of a Number

Description: Write a Java program to find the factorial of a number entered by the user.

Hint:

- Use a `for` loop to multiply numbers from 1 to the given number.
- Consider using recursion as an alternative approach.

6) Multiply Two Numbers in Java

Description: Write a Java program to multiply two numbers entered by the user and display the result.

Hint:

- Similar to the addition problem, use the `Scanner` class to take input.
- Multiply the numbers using the `*` operator.

7) Check Leap Year in Java

Description: Write a Java program to check if a given year is a leap year.

Hint:

- A year is a leap year if it is divisible by 4, but not by 100 unless it is also divisible by 400.
- Use if-else statements to implement the logic and display the result.

These assignments cover fundamental programming concepts and should help students build a strong foundation in Java.

8) Reverse a Number

Description: Write a Java program that takes a number as input and prints its reverse.

Hint:

- Use a `while` loop to extract the digits of the number.
- Construct the reversed number by multiplying the current result by 10 and adding the extracted digit.

9) Check Palindrome String

Description: Write a Java program to check if a given string is a palindrome (a string that reads the same forwards and backwards).

Hint:

- Use a `for` loop to compare characters from the beginning and end of the string moving towards the center.
- Alternatively, you can reverse the string and compare it with the original.

10) Sum of Digits

Description: Write a Java program that takes a number as input and calculates the sum of its digits.

Hint:

- Use a `while` loop to extract each digit of the number using the modulus operator (`%`).
- Add each extracted digit to a sum variable.
- Use integer division to reduce the number after each digit is extracted.